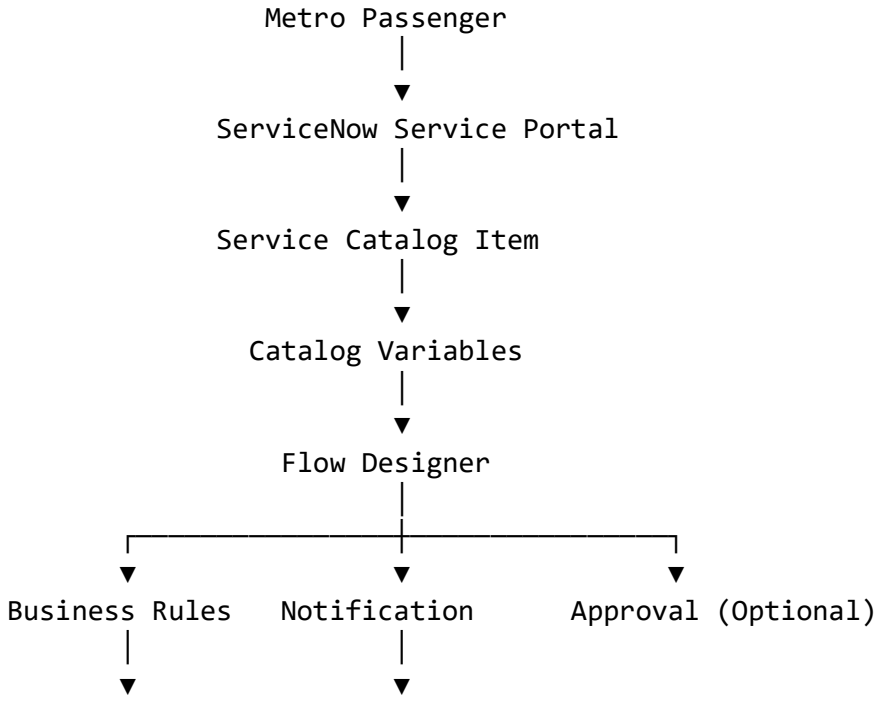
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 June 2026
Roll No	23A51A42A3
Project Name	Metro Ticket Generating System using ServiceNow
Maximum Marks	4 Marks

Technical Architecture



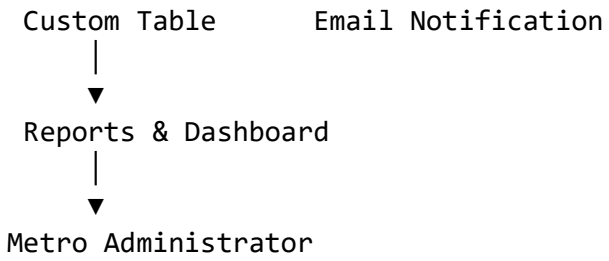


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Users book metro tickets through the Service Portal	**ServiceNow Service Portal**
2	Application Logic-1	Metro ticket booking process	**Service Catalog**
3	Application Logic-2	Ticket automation and workflow	**Flow Designer**
4	Application Logic-3	Data validation and processing	**Business Rules & Client Scripts**
5	Database	Stores passenger and booking details	**ServiceNow Custom Tables**
6	Cloud Database	Platform data storage	**ServiceNow Cloud Database**
7	File Storage	Stores attachments if required	**ServiceNow Attachments**
8	External API-1	Not Used	**N/A**
9	External API-2	Not Used	**N/A**
10	Machine Learning Model	Not Used	**N/A**
11	Infrastructure (Server / Cloud)	Cloud-hosted application	**ServiceNow Cloud Platform**

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology

1	Open-Source Frameworks	The project is developed entirely on the ServiceNow platform without additional open-source frameworks.	ServiceNow Platform
2	Security Implementations	Role-based access control (RBAC), user authentication, Access Control Lists (ACLs), and secure login protect application data.	ServiceNow ACLs, Roles
3	Scalable Architecture	The cloud-based architecture allows the application to support increasing numbers of users and ticket bookings.	ServiceNow Cloud Platform
4	Availability	The application is available through the ServiceNow cloud environment with high availability.	ServiceNow SaaS
5	Performance	Ticket booking requests are processed efficiently using Service Catalog, Flow Designer, and optimized database operations.	Service Catalog, Flow Designer